ASSIGNMENT

INTRODUCTION TO SYMMETRIC CRYPTOGRAPHY

Cryptography and Network Security

Instructions: All 8 questions are answered in simple, assignment-ready English. Diagrams are included where required.

Note: These answers follow the topics and terminology in the supplied assignment sheet. Write your name, roll number, class and submission date on the first page if required by your college.

Q1. Explain the basic principles of symmetric cryptography. Describe the roles of plaintext, ciphertext, secret key, encryption algorithm and decryption algorithm, and explain Kerckhoffs's principle. (10 Marks)

Symmetric cryptography is an encryption method in which the sender and receiver use the same secret key for encryption and decryption. It is used to protect information from unauthorized access. The sender converts readable data into an unreadable form, and the receiver uses the shared key to recover the original data.

Main components:

• **Plaintext:** The original readable message or data that needs protection.

• **Ciphertext:** The encrypted, unreadable form of plaintext.

• **Secret key:** A confidential value shared by the sender and receiver. The security of the system depends mainly on keeping this key secret.

• **Encryption algorithm:** The mathematical process that uses the plaintext and secret key to produce ciphertext.

• **Decryption algorithm:** The reverse process that uses the same secret key to convert ciphertext back into plaintext.

Working: First, the sender takes plaintext and applies the encryption algorithm with the secret key. The resulting ciphertext is sent through the communication channel. The receiver applies the decryption algorithm with the same secret key and obtains the original plaintext.

```
Plaintext  +  Secret Key\n        |\n        v\n  [Encryption Algorithm]\n        |\n        v\n     Ciphertext
```

Kerckhoffs's Principle: This principle states that a cryptographic system should remain secure even if its design or algorithm is publicly known. Only the secret key must be kept confidential. Therefore, security should not depend on hiding the algorithm. This makes a system easier to test and trust.

Example: AES is a symmetric encryption algorithm. The same secret key is used to encrypt and decrypt the data.

Q2. Distinguish between stream ciphers and block ciphers. Discuss confusion, diffusion and modes ECB, CBC, CFB, OFB and CTR with strengths and weaknesses. (10 Marks)

A **stream cipher** encrypts data one bit or one byte at a time by combining it with a generated keystream. A **block cipher** encrypts fixed-size blocks of data, such as 128-bit blocks in AES.

Stream cipher vs block cipher:

Stream cipher	Block cipher
Processes data bit-by-bit or byte-by-byte.	Processes fixed-size blocks of data.
Suitable for continuous data streams.	Suitable for files, messages and stored data.
Usually has low delay.	May need padding in modes that require complete blocks.
Example: RC4 (historical; now considered insecure).	Examples: DES and AES.

Confusion: It hides the relationship between the secret key and ciphertext. Substitution operations are commonly used to create confusion.

Diffusion: It spreads the effect of one plaintext bit across many ciphertext bits. Permutation and mixing operations help provide diffusion.

Block cipher modes of operation:

• **ECB (Electronic Codebook):** Each block is encrypted independently. *Strength:* Simple and allows parallel processing. *Weakness:* Identical plaintext blocks produce identical ciphertext blocks, revealing patterns; it is generally unsuitable for messages with repeated patterns.

• **CBC (Cipher Block Chaining):** Each plaintext block is XORed with the previous ciphertext block before encryption. The first block uses an initialization vector (IV). *Strength:* Hides repeated plaintext patterns when used correctly.

Weakness: Encryption is sequential and an error can affect a full block and the next block; padding is required.

• **CFB (Cipher Feedback):** The previous ciphertext is encrypted and combined with plaintext to create ciphertext. *Strength:* Can operate on smaller units and behaves like a self-synchronizing stream cipher. *Weakness:* Encryption is sequential and transmission errors can affect following data.

• **OFB (Output Feedback):** Repeated encryption of an IV produces a keystream, which is XORed with plaintext. *Strength:* Transmission errors do not spread beyond the affected bit. *Weakness:* IV/nonce reuse is dangerous, and the sender and receiver must stay synchronized.

• **CTR (Counter):** The cipher encrypts successive counter values to create a keystream, which is XORed with plaintext. *Strength:* Fast, parallelizable and supports random access. *Weakness:* A counter/nonce must never be reused with the same key.

Thus, the choice of mode depends on the need for speed, parallel processing, error handling and safe IV/nonce management.

Q3. Trace the historical development of symmetric encryption algorithms. (10 Marks)

Symmetric encryption has developed from simple manual methods into strong computer-based algorithms.

• **Caesar cipher:** One of the earliest substitution ciphers. Each letter is shifted by a fixed number of positions in the alphabet. It was easy to use but could be broken by trying the limited possible shifts.

• **Enigma machine:** Used by Germany during World War II. It used rotating electrical rotors to create changing letter substitutions. Its messages were difficult to break, but weaknesses in operating procedures and cryptanalysis helped Allied codebreakers read many messages.

• **Claude Shannon:** In the 1940s, Shannon applied information theory to secrecy systems. He explained important ideas such as confusion and diffusion, which influenced the design of modern ciphers.

• **Feistel cipher:** The Feistel structure divides a data block into two halves and processes them through repeated rounds. It became an important design approach for block ciphers.

• **DES in 1977:** The Data Encryption Standard was adopted as a U.S. federal standard in 1977. It used a 64-bit block and an effective 56-bit key. It was widely used, but its short key later became a major weakness.

• **Triple DES (3DES):** Triple DES applied DES three times to improve security and extend the useful life of DES. It was more secure than single DES but slower.

• **Other algorithms:** IDEA and Blowfish were developed as alternative symmetric block ciphers. RC4 was a widely used stream cipher, but later weaknesses made it unsuitable for secure modern use.

• **AES competition:** As DES became insecure against brute-force attacks, NIST organized an open competition to select a replacement. Rijndael was selected and became the Advanced Encryption Standard (AES) in 2001.

Conclusion: The history of symmetric encryption shows a move from simple letter substitution to carefully designed algorithms that use strong keys, multiple rounds, confusion and diffusion. AES is the widely adopted modern standard.

Q4. Explain Caesar, monoalphabetic, Playfair and Vigenere ciphers. Why is frequency analysis effective against some but not others? (10 Marks)

Classical substitution ciphers replace plaintext letters with other letters. Different methods use different rules.

• **(a) Caesar cipher:** Each letter is shifted by a fixed number of places. For a shift of 3, A becomes D, B becomes E and so on. *Strength:* Very simple. *Weakness:* Only 25 useful shifts exist, so brute force and frequency analysis can break it easily.

• **(b) Monoalphabetic substitution cipher:** Each plaintext letter is always replaced by a fixed different letter according to a substitution alphabet. *Strength:* Has many possible keys compared with Caesar. *Weakness:* Letter-frequency patterns remain, so frequency analysis can identify common letters and patterns.

• **(c) Playfair cipher:** It encrypts pairs of letters (digraphs) using a 5×5 key square. The pair is encrypted according to whether the letters are in the same row, same column or form a rectangle. I and J are commonly placed in one cell. *Strength:* Hides single-letter frequencies better than simple substitution. *Weakness:* Digraph patterns and language statistics can still reveal information.

• **(d) Vigenere cipher:** It uses a repeated keyword to apply different Caesar shifts to different letters. *Strength:* It changes the substitution from position to position and is stronger than a simple monoalphabetic cipher. *Weakness:* If the key repeats, its length can be estimated and frequency analysis can then be applied to groups of letters.

Frequency analysis: In English, some letters such as E, T and A occur more often than others. In Caesar and monoalphabetic substitution, each plaintext letter always maps to the same ciphertext letter, so the frequency pattern is preserved. Playfair works on pairs, so single-letter frequency is less useful. Vigenere uses multiple shifts, spreading the frequency pattern; however, a repeated key can create repeating patterns that help an attacker discover the key length.

Therefore, frequency analysis is most direct against simple substitution ciphers and more difficult, but not impossible, against digraph and repeating-key methods.

Q5. Describe rail fence and columnar transposition, product ciphers and one-time pad. (8 Marks)

A **transposition cipher** changes the positions of plaintext characters without changing the characters themselves.

• **Rail Fence cipher:** The message is written diagonally across a fixed number of rows (rails) and then read row by row. This rearranges the letters. It is simple but can be broken by trying different rail counts and examining language patterns.

• **Columnar Transposition cipher:** The plaintext is written in rows under a keyword. The columns are then read in the order determined by the alphabetical order of the keyword letters. It rearranges character positions but keeps all original letters. It is stronger than a basic rail fence cipher but still vulnerable to analysis.

Product cipher: A product cipher combines substitution and transposition. Substitution changes the values of symbols, while transposition changes their positions. Repeating these operations creates stronger protection. Modern block ciphers use repeated rounds of substitution and permutation/mixing, following this general idea.

One-Time Pad (OTP): It uses a truly random key that is as long as the message. Each plaintext bit is combined with the corresponding key bit (commonly using XOR). The key must be used only once and kept secret. If these conditions are perfectly followed, the OTP provides perfect secrecy and is theoretically unbreakable.

Why OTP is impractical: A separate random key as long as every message must be securely generated, delivered and stored. Reusing a key can destroy its security. Managing such long keys is difficult, especially for large amounts of data.

Q6. Describe the structure of DES, including Feistel structure, sizes, permutations, rounds, round function and key schedule. (12 Marks)

The **Data Encryption Standard (DES)** is a symmetric block cipher based on a Feistel structure. It encrypts data in 64-bit blocks using an effective key length of 56 bits. The key is often supplied as 64 bits, but 8 bits are used for parity.

Overall structure: The 64-bit plaintext first passes through an Initial Permutation (IP). It is divided into two 32-bit halves, called L0 and R0. These halves pass through 16 Feistel rounds. After the rounds, the halves are swapped and passed through the Final Permutation (IP■[1]) to produce 64-bit ciphertext.

```
64-bit Plaintext\n      |\n [Initial Permutation]\n      |\n  L0 (32 bits) | R0 (32 bits)\n      |\n [16 Feistel
```

DES round function: In each round, the right half is processed by function F with a 48-bit round key. The steps are:

• **Expansion:** The 32-bit right half is expanded to 48 bits.

• **Key mixing:** The expanded data is XORed with the 48-bit round key.

• **S-box substitution:** The 48-bit result is divided into eight 6-bit groups. Eight S-boxes convert them into eight 4-bit outputs, giving 32 bits.

• **P-box permutation:** The 32-bit S-box output is rearranged using a fixed permutation.

Feistel operation: For round i, the new left half is the previous right half, and the new right half is the previous left half XORed with F(previous right half, round key). In simple form: L■ = R■■■ and R■ = L■■■ XOR F(R■■■, K■).

```
One DES Round\nL(i-1) ---------------------> XOR -----------------> R(i)\n                                                    ^\nR(i-1
```

Key schedule: The 56 effective key bits are rearranged using Permuted Choice 1 (PC-1) and divided into two 28-bit halves. The halves are shifted left by one or two positions in each round. Permuted Choice 2 (PC-2) selects 48 bits to form each round key. This produces 16 round keys, K1 to K16.

Decryption: DES uses the same Feistel structure for decryption, but the round keys are applied in reverse order, from K16 to K1.

Q7. Discuss DES strengths and weaknesses, Double DES and Triple DES. (10 Marks)

DES was an important and influential block cipher. Its 16-round Feistel structure and S-boxes provide useful security properties, but its effective key length is now too short.

Strengths of DES:

• **Avalanche effect:** A small change in plaintext or key causes many ciphertext bits to change. This makes the relationship between input and output difficult to predict.

• **Structured design:** Sixteen rounds, expansion, S-boxes and permutations provide confusion and diffusion.

Weaknesses of DES:

• **Short key:** DES has only a 56-bit effective key, giving 2■■ possible keys. Modern computers can search this key space using brute force.

• **Brute-force attack:** An attacker tries possible keys until meaningful plaintext is found. The short key makes this practical with sufficient computing resources.

• **Differential cryptanalysis:** It studies how differences in chosen plaintext pairs affect differences in ciphertext. It is a method for analyzing block ciphers.

• **Linear cryptanalysis:** It uses approximate linear relationships among plaintext, ciphertext and key bits to gain information about the key.

Double DES: It applies DES encryption twice with two keys: C = E(K2, E(K1, P)). It appears to provide a longer key, but it is vulnerable to a **meet-in-the-middle attack**. The attacker encrypts the plaintext with all possible K1 values and decrypts the ciphertext with all possible K2 values, then looks for a matching intermediate value. This reduces the expected security improvement.

Triple DES (3DES): It applies DES three times, commonly in Encrypt-Decrypt-Encrypt (EDE) form. It was introduced to improve DES security. Two-key 3DES uses two independent keys (with K1 repeated as the third key), while three-key 3DES uses three independent keys. Three-key 3DES offers a larger key space than two-key 3DES, but both are slower than modern AES.

Conclusion: DES is no longer suitable for protecting new sensitive data because of its short key. Triple DES improved its security but is being phased out in favor of AES.

Q8. Explain AES, its block/key sizes, rounds, transformations, key expansion and compare it with DES. (10 Marks)

The **Advanced Encryption Standard (AES)** is a symmetric block cipher selected through an open competition organized by NIST. It is based on the Rijndael algorithm and is widely used to protect digital information.

Block and key sizes: AES always uses a 128-bit block. It supports three key lengths:

Key length	Number of rounds
128 bits	10
192 bits	12

Key length	Number of rounds
256 bits	14

Four round transformations:

• **SubBytes:** Each byte is replaced using a substitution table called the S-box. It provides confusion.

• **ShiftRows:** The rows of the state matrix are shifted by different offsets. It rearranges bytes and supports diffusion.

• **MixColumns:** Bytes in each column are mathematically combined to spread their influence across the column. This provides diffusion.

• **AddRoundKey:** The state is combined with a round key using XOR.

Working of AES: The 128-bit plaintext is arranged as a 4×4 matrix of bytes called the state. First, an initial AddRoundKey operation is performed. Then the main rounds apply SubBytes, ShiftRows, MixColumns and AddRoundKey. In the final round, MixColumns is omitted. The number of rounds depends on the key length.

Key expansion: The original key is expanded to generate a separate round key for each round. The process uses operations such as RotWord, SubWord and round constants (Rcon).

Decryption: AES decryption applies the inverse transformations in reverse order: InvShiftRows, InvSubBytes, AddRoundKey and InvMixColumns. The final round follows the corresponding inverse sequence without InvMixColumns. The round keys are used in reverse order.

AES vs DES:

Feature	DES	AES
Structure	16-round Feistel network	Substitution-permutation network
Block size	64 bits	128 bits
Key size	56 effective bits	128, 192 or 256 bits
Rounds	16	10, 12 or 14
Security	Insecure for modern protection due to short key	Strong when correctly implemented with suitable key management
Performance	Older design; 3DES is relatively slow	Generally faster and more efficient in modern systems

Conclusion: AES provides stronger security and better modern performance than DES. It is the preferred standard for modern symmetric encryption.